

Department of Computer and Software Engineering**SE: Machine Learning**

Course Instructor: Dr. Ahmed Raza	Date: 05-March-2026
Day: Thursday	Semester: 6th

Lab 4: Hard vs Soft Margin using C, Hinge Loss and Kernels

Lab Title	CLO	BT	PLO	Weightage
Soft Margin Behaviour using C and Hinge Loss				

Name	Roll Number	Report /10	Viva /5	Total /15
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Signature: _____

1 Learning Outcomes

After completing this lab, the student will be able to:

- Understand the role of parameter C in SVM.
- Relate large C to hard-margin behaviour.
- Analyze robustness of soft-margin SVM under noise.
- Compute hinge loss as violation measure.
- Observe kernel-based nonlinear separation.

2 Datasets

Students will work with three deterministic datasets:

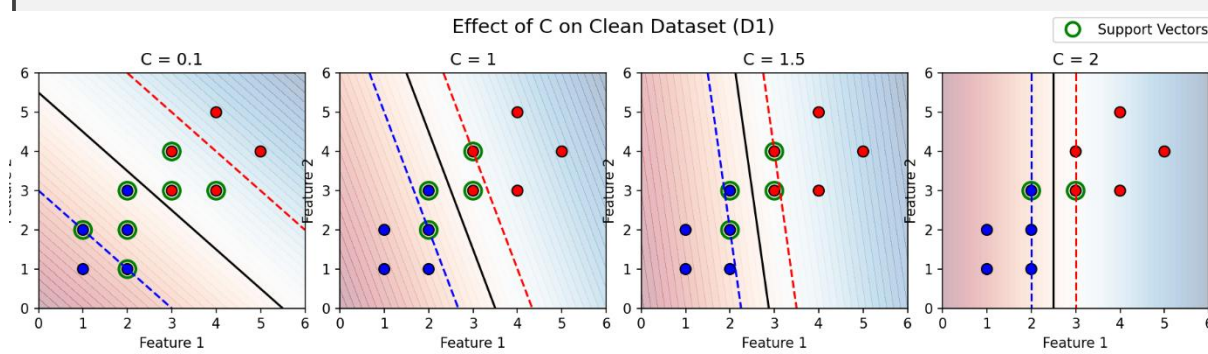
- D1: Clean Separable
- D2: Overlapping
- D3: Noisy (2 misclassified points)

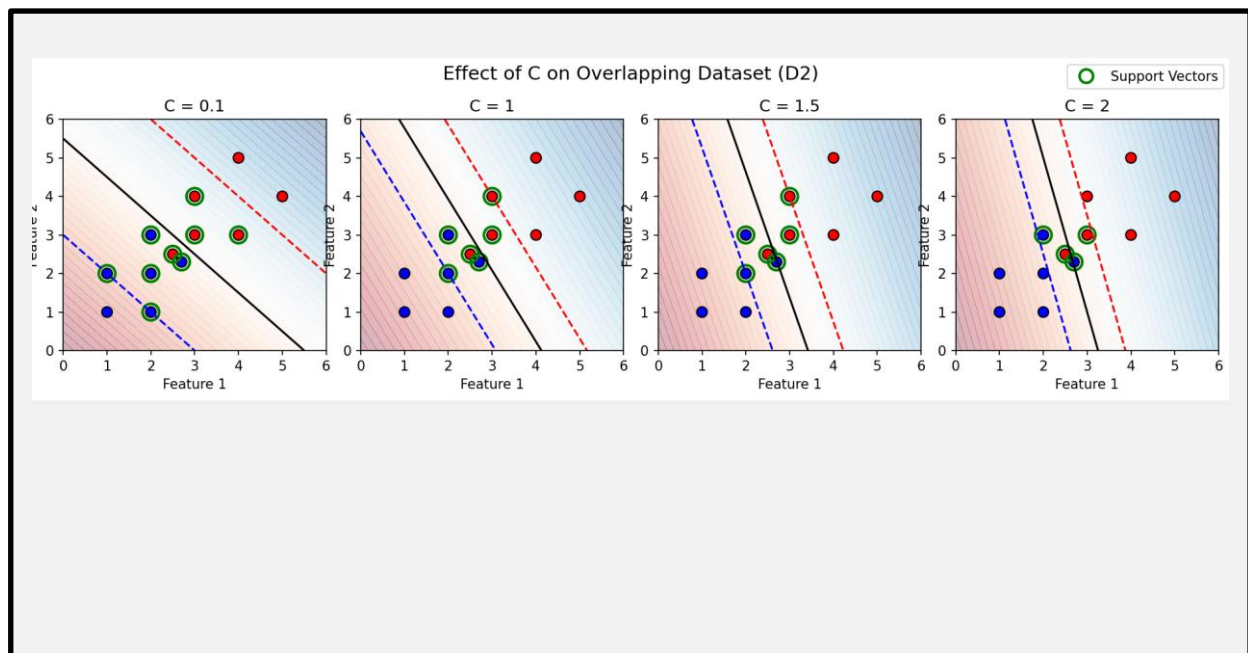
3 Part A: Effect of C (4 Marks)

Using D1 and D2, train SVM with:

- $C = 0.1$
- $C = 1$
- $C = 1.5$
- $C = 2$

Plot decision boundary and margins.





Observation

Comment on how boundary changes with C.
Does large C behave like hard-margin?

- When C is small, the SVM keeps a wide margin and does not care much if some points are on the wrong side. As we increase C, the model tries harder to get every point right, so the margin gets smaller and the boundary moves closer to the data points.
- When C is very large, it works almost like hard-margin SVM because it does not want any mistakes at all. This is fine when the data is clean, but when there is overlap between classes, it forces the boundary too close to the points which is not ideal.

4 Part B: Noise Sensitivity (3 Marks)

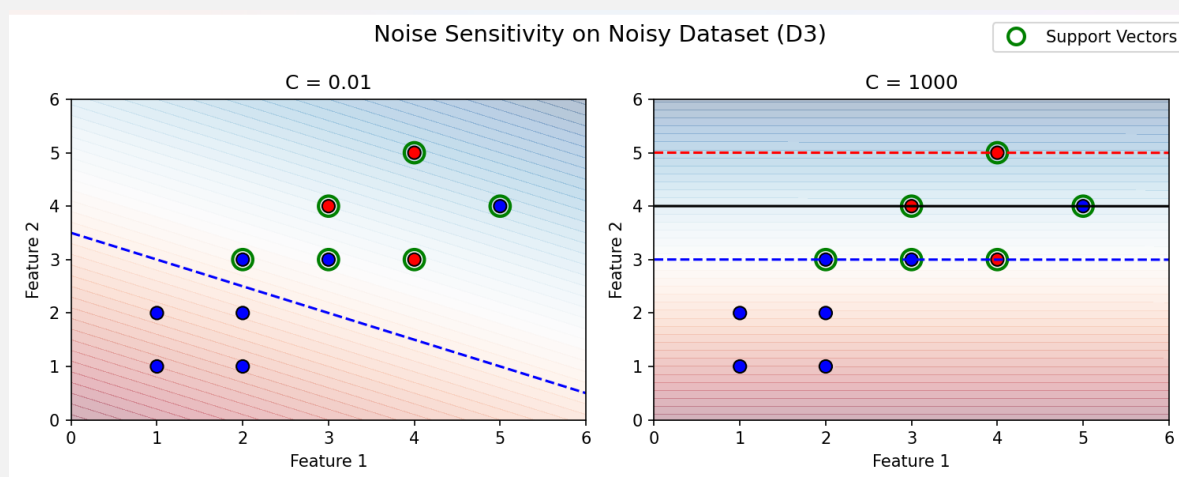
Using D3:

Train with:

- Very large C
- Small C

Compare

boundaries.



Comparison

Small C ($C = 0.01$) $\rightarrow$ Soft Margin

- The boundary we get is smooth and does not try too hard to get every point right. It just looks at the general pattern in the data.
- The 2 mislabeled points do not really affect the boundary because the model is okay with some mistakes happening.
- We see more support vectors here because the margin is kept wide and some points are allowed to be inside the margin.
- Overall it handles noise well since the wrong labels do not mess up the decision boundary.

Very Large C ($C = 1000$) $\rightarrow$ Hard Margin-like

- The boundary moves a lot and tries to fit every single point correctly even the ones with wrong labels.
- It treats the flipped labels as if they are correct and adjusts the boundary to fit them which causes overfitting.
- There are fewer support vectors but the boundary does not look natural because it is being forced to fit noisy points.
- Even just 2 wrong labels change the boundary a lot so the model is very sensitive to noise.

Part C: Understanding Hinge Loss and its Connection with C (4 Marks)

So far, you have used SVC as a black-box model and observed how the parameter C affects the decision boundary.

Now we dig slightly deeper.

Recall that SVM does NOT minimize classification error directly.

Instead, it minimizes:

$$\frac{1}{2}||w||^2 + C \sum_{i=1} L_i \quad (1)$$

Where hinge loss for a sample is:

$$L_i = \max(0, 1 - y_i(w^T x_i + b)) \quad (2)$$

This tells us:

- $||w||^2 \rightarrow$ controls margin width (simplicity of model)
- Hinge loss $\rightarrow$ penalizes margin violations
- $C \rightarrow$ controls how much we care about hinge loss

- Implement Hinge Loss.
 - Write a function that computes hinge loss for a dataset.
 - Train SVM multiple times with ($C = [0.01, 0.1, 1, 10, 100]$) using sklearn and extract W and b using
$$\text{decision_values} = \text{model.decision_function}(X)$$
 - Compute total hinge loss for all
 - Plot decision boundary
 - Count number of support vectors
 - Create a table with columns C , Total HingeLoss and no. of support vectors

Code

```
def hinge_loss(X,y,w,b):  
    """  
    Compute total hinge loss for a linear model.  
  
    Instructions for students:  
    - For each sample (x_i, y_i), compute:  
    |     loss_i = max(0, 1 - y_i * (w^T x_i + b))  
    - Sum over all samples to get total hinge loss.  
    - Return the total loss.  
    - Do NOT include any sklearn functions here; use  
    only numpy.  
    """  
    # calculating hinge loss for each point  
    total_loss = 0  
    num_samples = X.shape[0]  
  
    for i in range(num_samples):  
        margin = y[i] * (np.dot(w, X[i]) + b)  
        sample_loss = max(0, 1 - margin)  
        total_loss = total_loss + sample_loss  
  
    return total_loss
```

Training

```
model_c = train_svm(Xoverlap,yoverlap,C=cval,
kernel='linear')
weights,b_val = get_w_b(model_c)
```

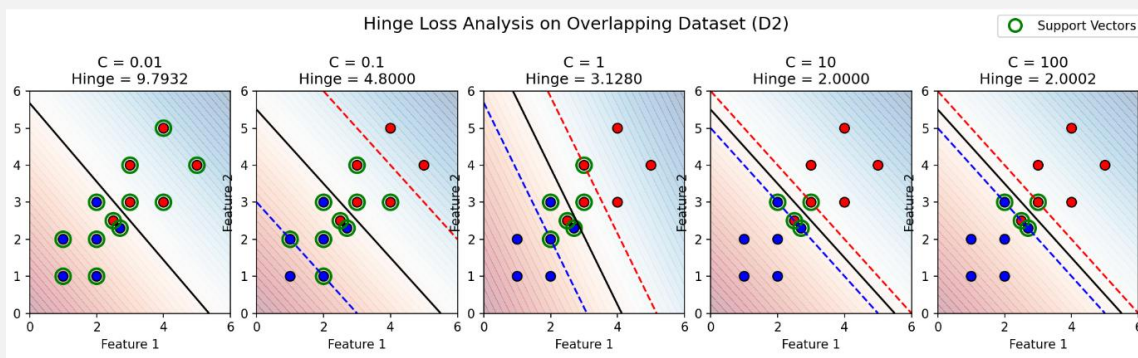
Total Hinge Loss For all + Support Vectors + Table

Hinge Loss Analysis on Overlapping Dataset

C	Hinge Loss	Support Vectors
0.01	9.7932	12
0.1	4.8000	9
1	3.1280	6
10	2.0000	4
100	2.0002	4

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Plots



- Observe Effect of C : Analysis Questions

1. What happens to hinge loss when C increases?
2. Why does hinge loss decrease with larger C ?
3. How does this relate to hard-margin behaviour?
4. When C is small, what happens to margin violations?

Answer1:

When we increase the value of C the hinge loss goes down. This is because the model tries harder to classify every point correctly. A bigger C means the model does not want to allow any mistakes so the loss becomes smaller and smaller.

Answer2:

Hinge loss decreases because a larger C puts more penalty on the misclassified points. The model is being told that making errors is very costly so it adjusts the decision boundary in a way that reduces these errors. It tries to fit the training data as closely as possible which makes the total hinge loss go down.

Answer3:

When C becomes very large the SVM starts behaving like a hard margin classifier. A hard margin SVM does not allow any point to be on the wrong side of the boundary. So when C is very high the model does the same thing. It pushes the boundary to perfectly separate the classes and the margin becomes very narrow. This means the model is very strict and does not tolerate any violations at all just like a hard margin SVM.

Answer4:

When C is small the model becomes more relaxed. It allows some points to be inside the margin or even on the wrong side of the boundary. These are called margin violations. A small C means the model does not care too much about individual mistakes and instead focuses on finding a wider margin. So more points end up violating the margin but the overall boundary is smoother and more general. This makes the model less sensitive to noise in the data.

1 Part D: Kernel Trick (4 Marks)

Generate concentric circle dataset. Train using:

- Linear kernel
- Polynomial kernel
- RBF kernel

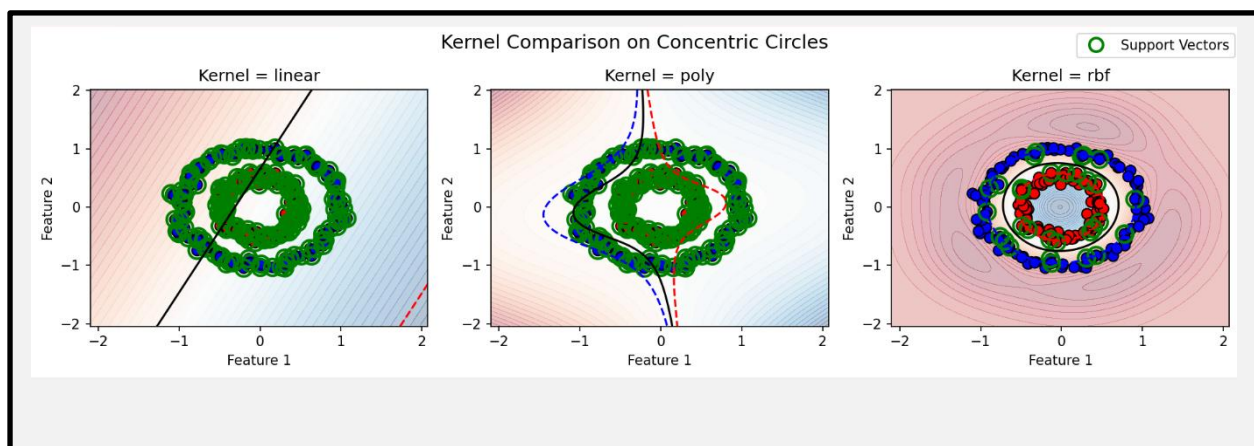
Plot decision boundaries.

Generating Datasets

```
def generate_circles():  
    from sklearn.datasets import make_circles  
    X,y = make_circles(n_samples=200,noise=0.05,factor=0.5,random_state=0)  
    y[y==0] = -1  
    return X,y
```

Training

```
# trying all three kernels on circle data  
Xcircles,ycircles = generate_circles()  
kernels = ['linear','poly','rbf']  
  
fig,axes = plt.subplots(1,len(kernels),figsize=(5*len(kernels),4))  
fig.suptitle('Kernel Comparison on Concentric Circles',fontsize=14)  
  
for j,ktype in enumerate(kernels):  
    plt.sca(axes[j])  
    model_d = train_svm(Xcircles,ycircles,C=1,kernel=ktype)  
    plot_decision_boundary(Xcircles,ycircles,model_d,f'Kernel = {ktype}')  
    legend_handle = Line2D([0],[0],marker='o',color='w',markeredgecolor='green',  
                           markerfacecolor='none',markersize=10,  
                           markeredgewidth=2,  
                           label='Support Vectors')  
    fig.legend(handles=[legend_handle],loc='upper right',fontsize=10)  
plt.tight_layout()  
plt.show()
```



Assessment Rubrics

Method: Lab reports and instructor observation during lab sessions

Outcome Assessed:

1. Ability to conduct experiments, as well as to analyze and interpret data (P).
2. Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice (P).

Performance	Exceeds expectation (4-5)	Meets expectation (3-2)	Does not meet expectation (1)	Marks
1. Realization of Experiment [a, b]	Selects relevant equipment to the experiment, develops setup diagrams of equipment connections or wiring.	Needs guidance to select relevant equipment and to develop equipment connection or wiring diagrams.	Incapable of selecting relevant equipment to conduct the experiment.	
2. Conducting Experiment [a, b]	Does proper calibration of equipment, carefully examines equipment moving parts, and ensures smooth operation.	Calibrates equipment, examines equipment moving parts, and operates the equipment with minor error.	Unable to calibrate appropriate equipment, and equipment operation is substantially wrong.	
3. Laboratory Safety Rules [a]	Respectfully and carefully observes safety rules and procedures.	Observes safety rules and procedures with minor deviation.	Disregards safety rules and procedures.	
6. Data Collection [a]	Plans data collection to achieve experimental objectives, and conducts an orderly and complete data collection.	Plans data collection to achieve experimental objectives, and collects complete data with minor error.	Does not know how to plan data collection; data collected is incomplete and contain errors.	
7. Data Analysis [a]	Accurately conducts simple computations and statistical analysis; correlates results to known theoretical values; accounts for measurement errors.	Conducts simple computations with minor error; reasonably correlates results to known theoretical values; attempts to account for errors.	Unable to conduct simple statistical analysis; no attempt to correlate or explain errors.	
8. Computer Use [a]	Uses computer to collect and analyze data effectively.	Uses computer to collect and analyze data with minor error.	Does not know how to use computer to collect and analyze data.	
Total				

Faculty:

Name: _____

Signature: _____

Date: _____